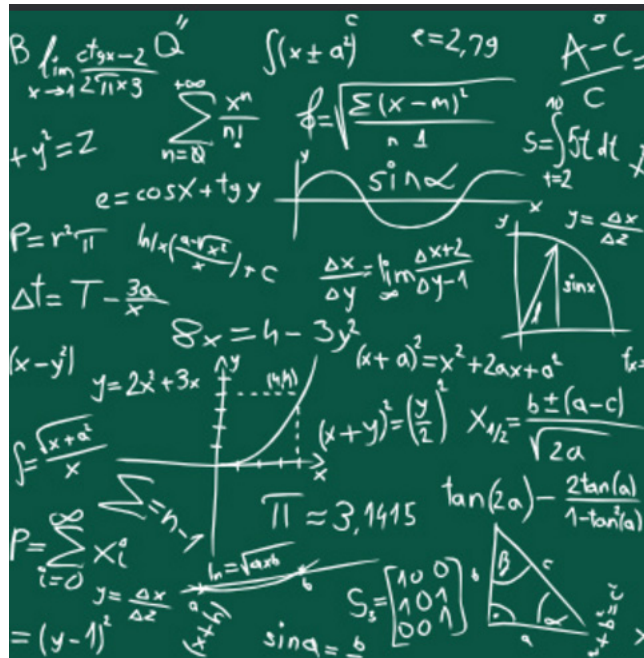


Power functions



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Contents

1	Power functions	2
1.1	Rational exponent	2
1.2	Real exponent	6
1.3	Polynomials	7
2	Direct and inverse proportionality functions	8

1 Power functions

Definition 1: power functions

Given $\alpha \in \mathbb{R}, \alpha \neq 0$, the function:

$$f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}, \quad f : x \mapsto x^\alpha \quad (1)$$

is called the **power function** with exponent α . The domain D depends on the value of α :

- with rational exponent $\alpha = \frac{m}{n} \in \mathbb{Q}$, $m \in \mathbb{Z}$ and $n \in \mathbb{N}_+$ coprime, we have four cases:

$$D = \begin{cases} \mathbb{R} & \text{if } n \text{ odd and } \frac{m}{n} > 0 \\ \mathbb{R} \setminus \{0\} & \text{if } n \text{ odd and } \frac{m}{n} < 0 \\ [0, +\infty) & \text{if } n \text{ even and } \frac{m}{n} > 0 \\ (0, +\infty) & \text{if } n \text{ even and } \frac{m}{n} < 0 \end{cases}$$

- with real, non-rational exponent $\alpha \in \mathbb{R} \setminus \mathbb{Q}$, we have two cases:

$$D = \begin{cases} [0, +\infty) & \text{if } \alpha > 0 \\ (0, +\infty) & \text{if } \alpha < 0 \end{cases}$$

1.1 Rational exponent

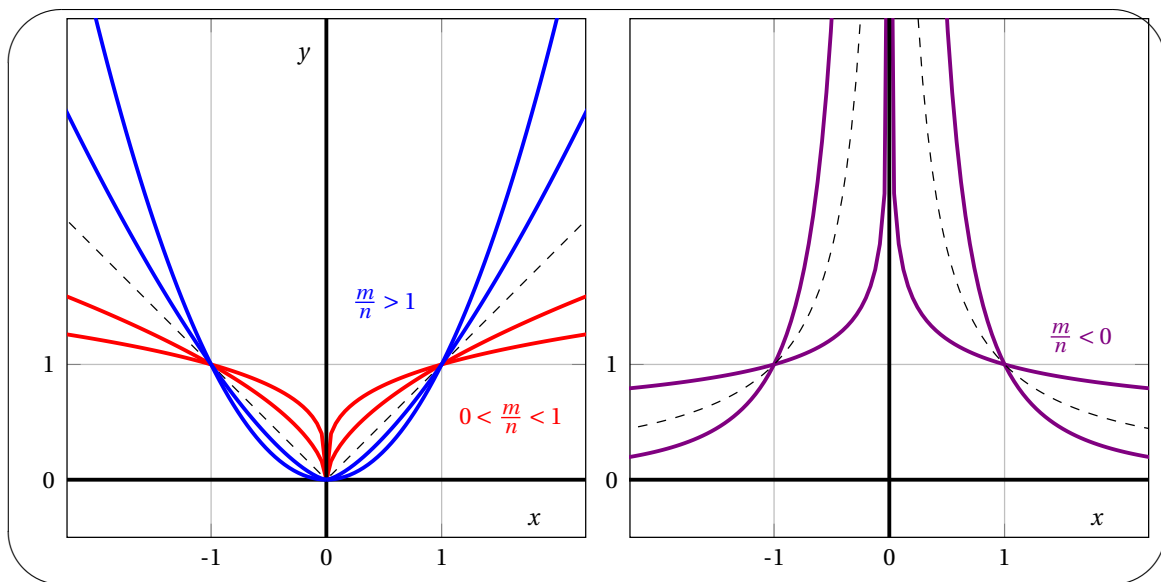
With $\alpha = \frac{m}{n} \in \mathbb{Q}$, $m \in \mathbb{Z}$ and $n \in \mathbb{N}_+$ coprime, we have a power function with rational exponent:

$$f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}, \quad f : x \mapsto x^{\frac{m}{n}} = \sqrt[n]{x^m}$$

We have the following three cases, depending on whether m and n are even or odd.

1. With n odd and m even, it is an **even function**, i.e., $(-x)^{\frac{m}{n}} = x^{\frac{m}{n}}$. For the sign and the monotonicity we have:

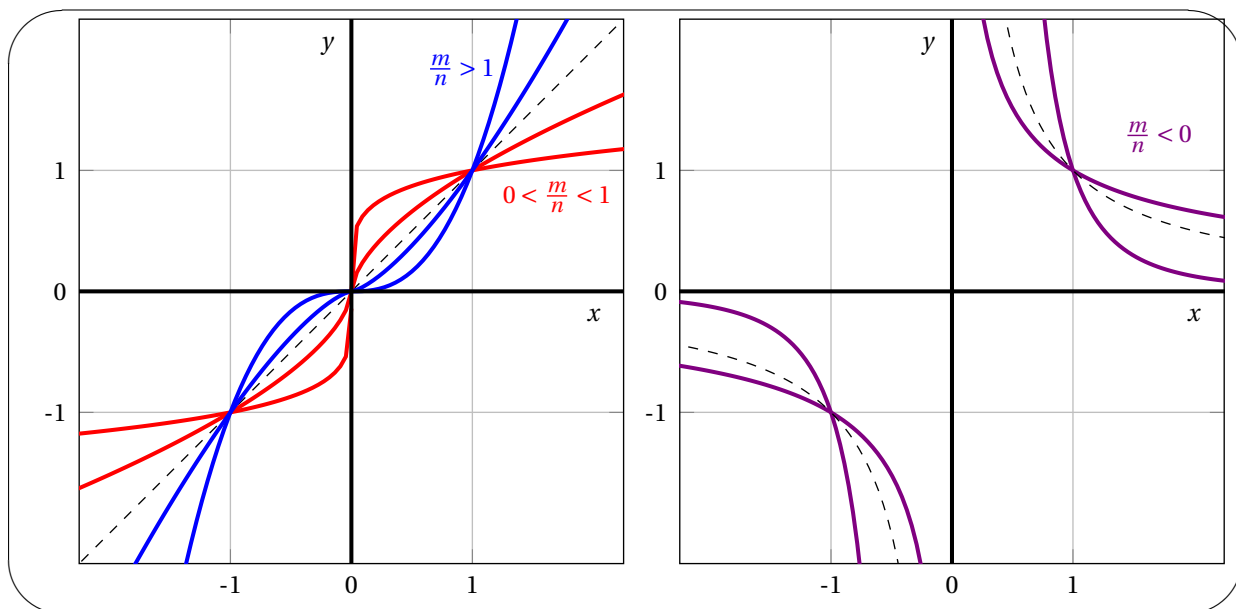
$$\begin{aligned} & x^{\frac{m}{n}} > 0, \quad \forall x \in \mathbb{R} \setminus \{0\}; \quad x^{\frac{m}{n}} = 0 \iff x = 0 \text{ and } \frac{m}{n} > 0 \\ & \begin{cases} \text{if } \frac{m}{n} > 0 \text{ then } f \text{ is decreasing in } (-\infty, 0] \text{ and increasing in } [0, +\infty) \\ \text{if } \frac{m}{n} < 0 \text{ then } f \text{ is increasing in } (-\infty, 0) \text{ and decreasing in } (0, +\infty) \end{cases} \end{aligned}$$



2. With n odd and m odd, it is an **odd function**, i.e., $(-x)^{\frac{m}{n}} = -x^{\frac{m}{n}}$. For the sign and the monotonicity we have:

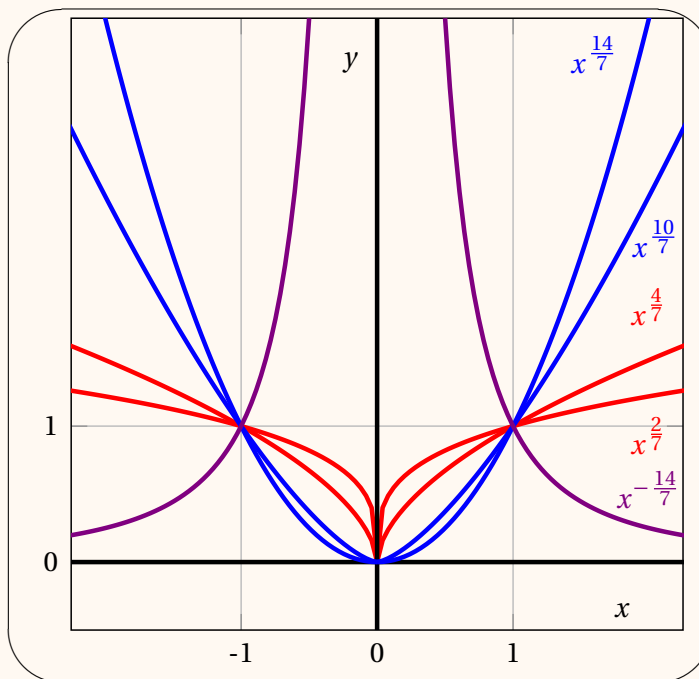
$$x^{\frac{m}{n}} < 0, \forall x < 0, \quad x^{\frac{m}{n}} > 0, \forall x > 0; \quad x^{\frac{m}{n}} = 0 \iff x = 0 \text{ and } \frac{m}{n} > 0$$

$$\begin{cases} \text{if } \frac{m}{n} > 0 \text{ then } f \text{ is increasing in } \mathbb{R} \\ \text{if } \frac{m}{n} < 0 \text{ then } f \text{ is decreasing in } (-\infty, 0) \text{ and in } (0, +\infty) \end{cases}$$

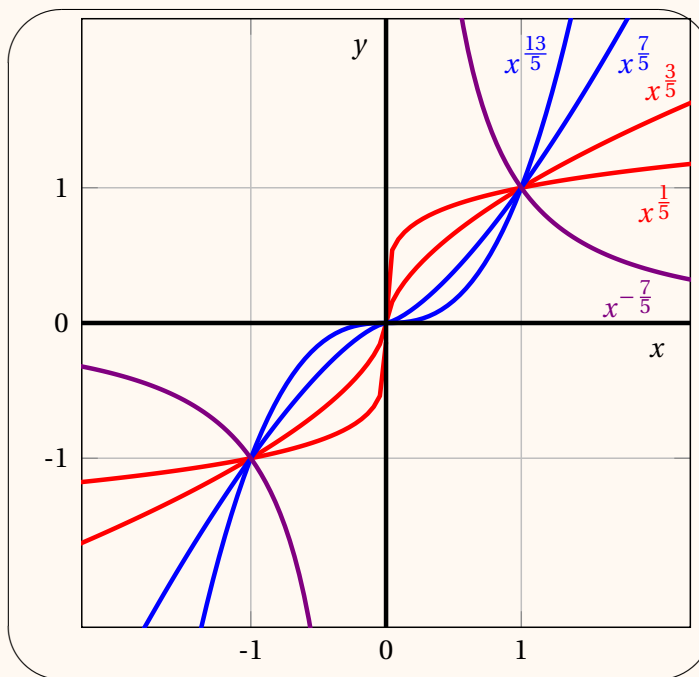


Example 1: graphs of power functions with rational exponent $\frac{m}{n}$ and n odd

With m even:



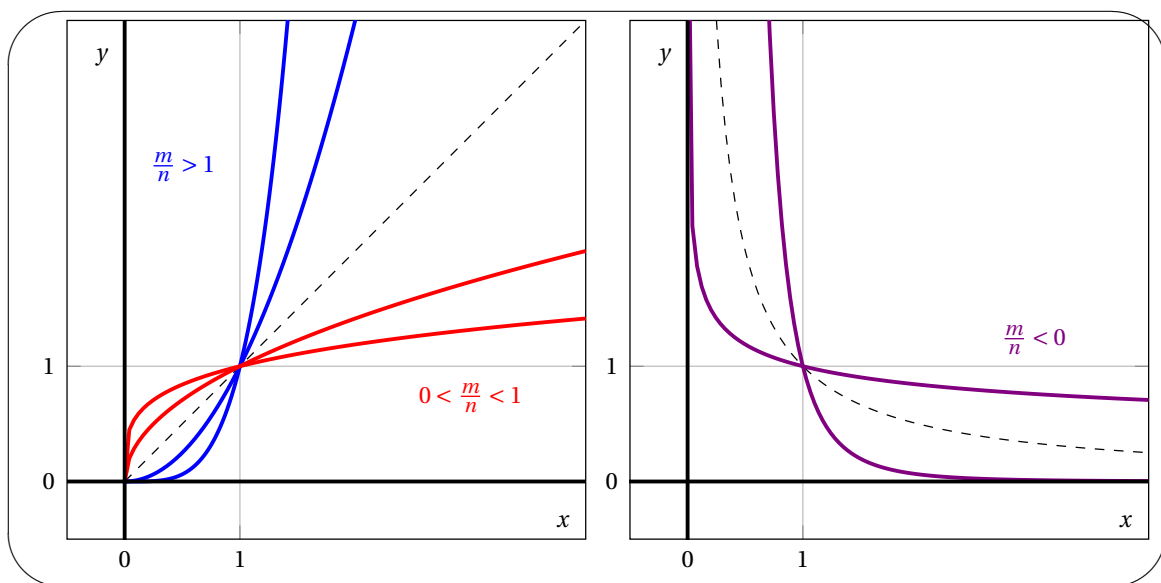
With m odd:



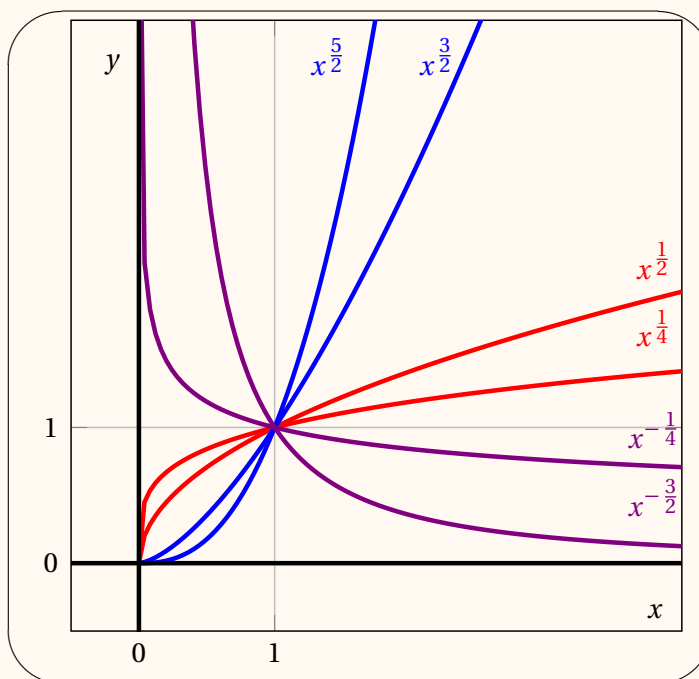
3. With n even and m odd (m cannot be even since they are coprime), for the sign and the monotonicity we have:

$$x^{\frac{m}{n}} > 0, \quad \forall x > 0; \quad x^{\frac{m}{n}} = 0 \iff x = 0 \text{ and } \frac{m}{n} > 0$$

$$\begin{cases} \text{if } \frac{m}{n} > 0 \text{ then } f \text{ is increasing in } [0, +\infty) \\ \text{if } \frac{m}{n} < 0 \text{ then } f \text{ is decreasing in } (0, +\infty) \end{cases}$$



Example 2: graphs of power functions with rational exponent $\frac{m}{n}$ and n even

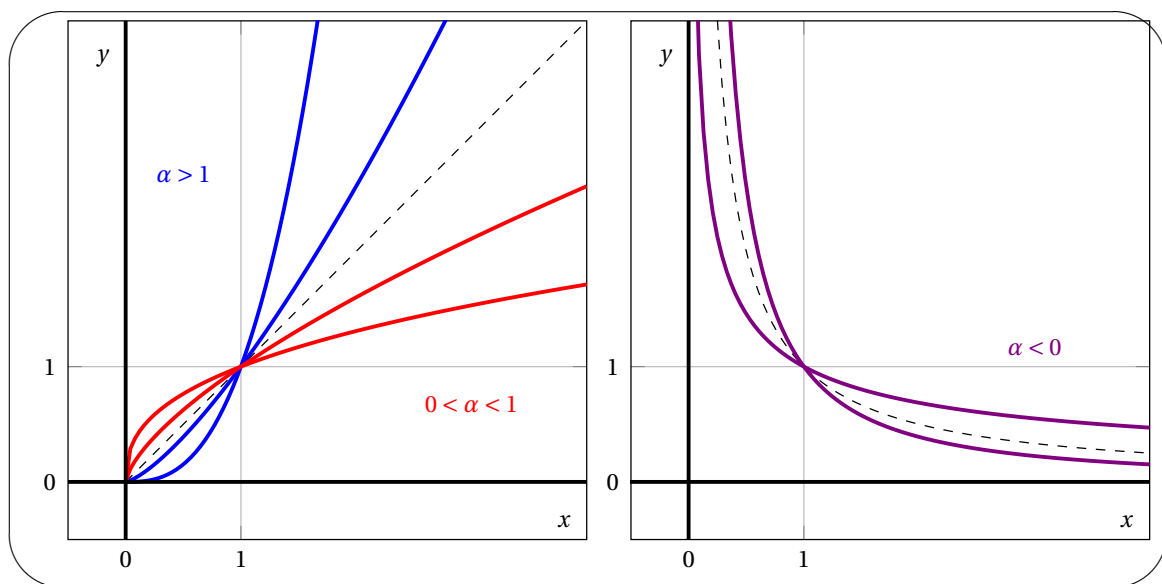


1.2 Real exponent

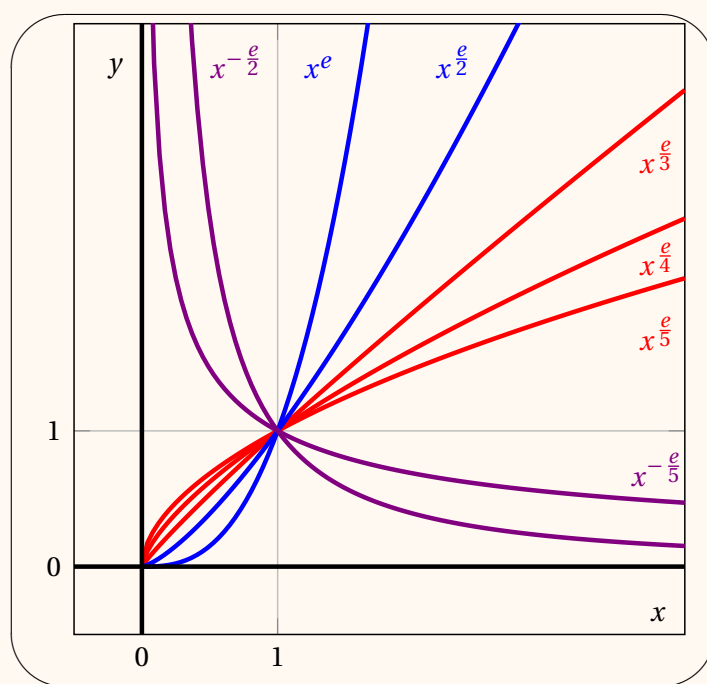
- With $\alpha \in \mathbb{R} \setminus \mathbb{Q}$, for the sign and the monotonicity we have:

$$x^\alpha > 0, \quad \forall x > 0; \quad x^\alpha = 0 \iff x = 0 \text{ and } \alpha > 0$$

$$\begin{cases} \text{if } \alpha > 0 \text{ then } f \text{ is increasing in } [0, +\infty) \\ \text{if } \alpha < 0 \text{ then } f \text{ is decreasing in } (0, +\infty) \end{cases}$$



Example 3: graphs of power functions with real exponent



1.3 Polynomials

Definition 2: polynomial

Given $n + 1$ values $a_i \in \mathbb{R}$ with $i \in \{0, 1, \dots, n\}$ and $a_n \neq 0$, the function:

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f: x \mapsto \underbrace{\sum_{i=0}^n a_i x^i}_{=P_n(x)} \quad (2)$$

is called a **polynomial** of degree n .

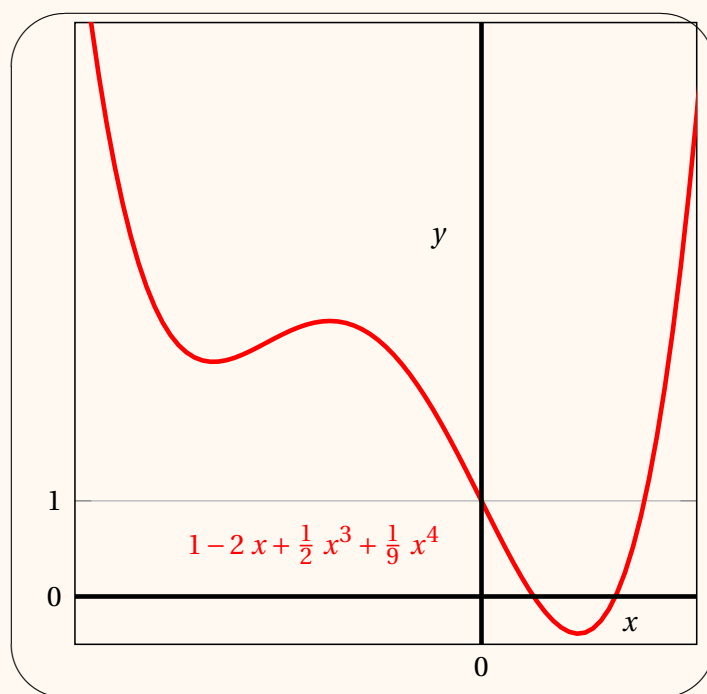
- For each **monomial** $i \in \{0, 1, \dots, n\}$:
 - the value a_i is the **coefficient of the monomial**
 - the power function x^i with integer exponent i is the **literal part** of the monomial

The value a_0 is the **constant term** of the polynomial since $x^0 = 1$.

Example 4: polynomial

For example, with $n = 4$, $a_0 = 1$, $a_1 = -2$, $a_2 = 0$, $a_3 = \frac{1}{2}$ and $a_4 = \frac{1}{9}$ we have the following polynomial of degree four:

$$P_4(x) = \sum_{i=0}^4 a_i x^i = 1 - 2x + \frac{1}{2}x^3 + \frac{1}{9}x^4$$

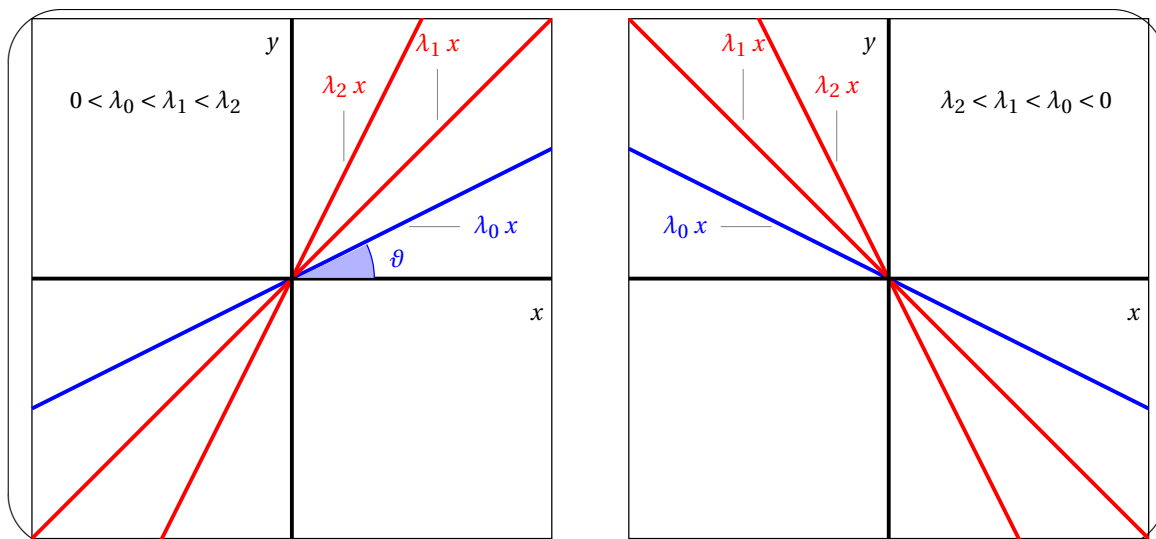


2 Direct and inverse proportionality functions

With $\alpha = 1$ and given $\lambda \in \mathbb{R}$, we have the family of **direct proportionality (or linear) functions**:

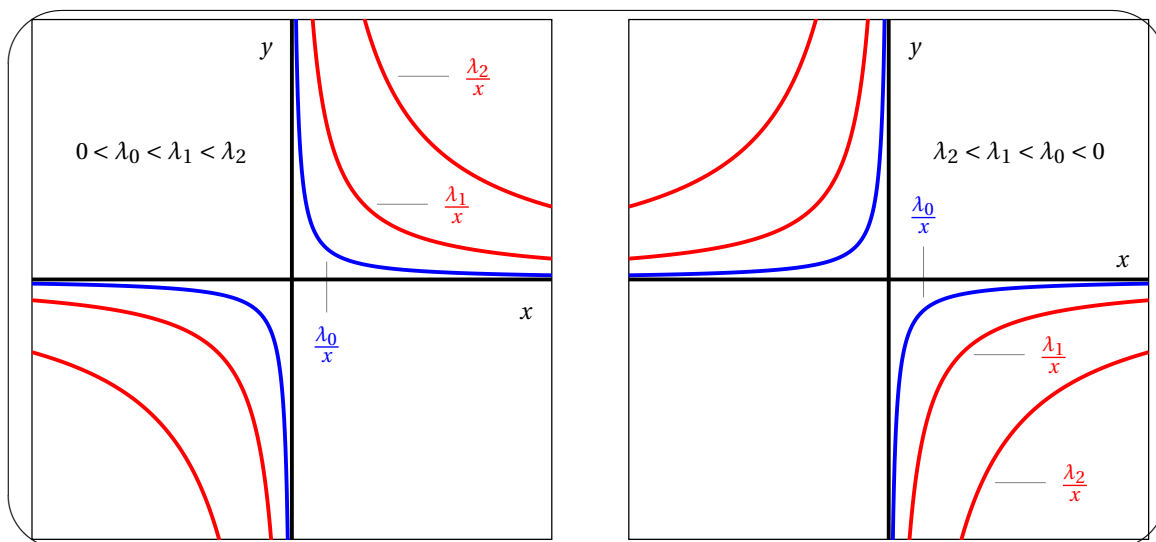
$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f: x \mapsto \lambda x \quad \text{where } \lambda \text{ is the constant of direct proportionality}$$

Moreover, $\lambda = \tan \vartheta$ and ϑ is the angle between the line $y = \lambda x$ and the x -axis.



With $\alpha = -1$ and given $\lambda \in \mathbb{R}$, we have the family of **inverse proportionality functions (or rectangular hyperbolas)**:

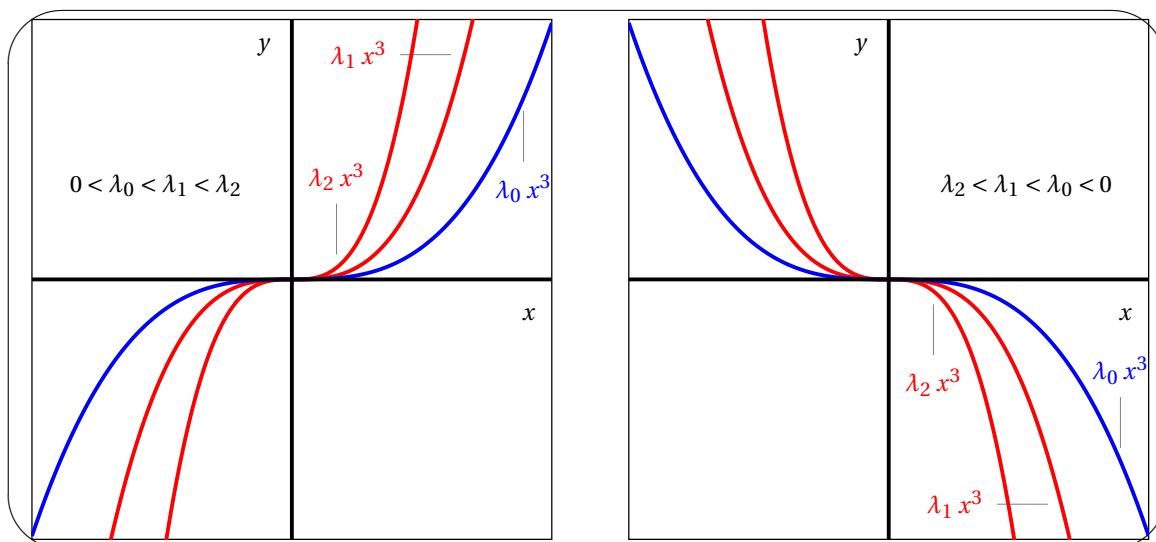
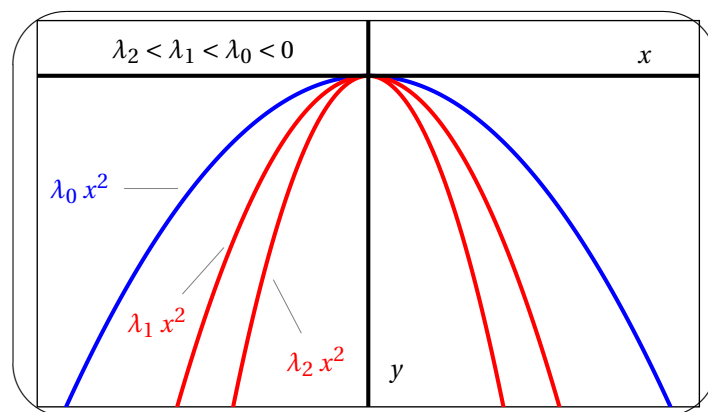
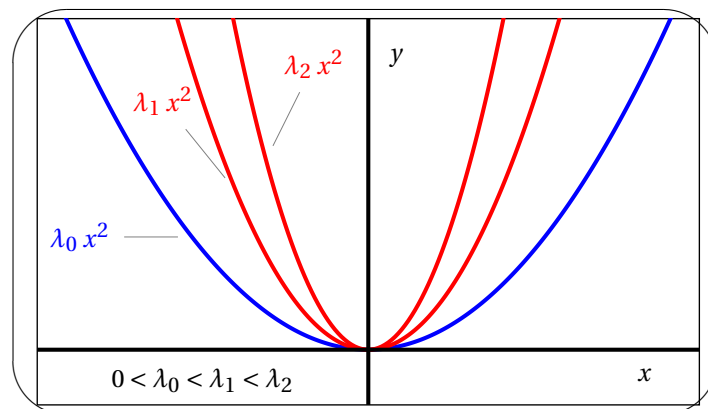
$$f: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}, \quad f: x \mapsto \frac{\lambda}{x} \quad \text{where } \lambda \text{ is the constant of inverse proportionality}$$



Given $\alpha \in \mathbb{R}$ and $\lambda \in \mathbb{R}$, we have the family of **power functions multiplied by a constant**:

$$f: D \subseteq \mathbb{R} \rightarrow \mathbb{R}, \quad f: x \mapsto \lambda x^\alpha \quad \text{where } \lambda \text{ is the multiplicative constant}$$

- For example, with α equal to 2 or equal to 3 and $\lambda \in \mathbb{R}$ we have:



- For example, with $\alpha = \frac{1}{2}$ and $\lambda \in \mathbb{R}$ we have:

